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Regulation: R22

Code No: 22PY104/2

I B. Tech I Semester Supplementary Examinations – May, 2023

PHYSICS FOR ELECTRONIC ENGINEERS

Time: 150 Min

(ECE)

Max. Marks: 80M

SECTION – A

Answer all Four questions

4×10M=40M

1. a) Analyze the following parameters for FCC and BCC lattices.
 - (i) Effective number of atoms (ii) Coordination number (iii) Void space [3M]
- b) How will you determine Miller indices of a lattice plane? Illustrate with an example. [4M]
- c) Find the ratio between $d_{(110)}$ and $d_{(111)}$ planes for a cubic system. [3M]
2. a) Obtain an expression for deBroglie's wavelength and express it in terms of kinetic energy and applied potential. [3M]
- b) Compare the wavelengths of electrons and neutrons both with kinetic energy 3 eV.
Given mass of electron – 9.1×10^{-31} Kg and neutron is 1.674×10^{-27} Kg respectively. [3M]
- c) An excited state of a certain nucleus has life-time of 5×10^{-18} s. Find the minimum possible uncertainty in its energy stating Heisenberg uncertainty principle. [4M]
3. a) How are electric potential and electric field related at a given point? [3M]
- b) Discuss the following cases for the intensity of electric field due to uniformly charged spherical shell with charge Q and radius R.
 - (i) Point P lies outside the shell at a distance r ($r > R$)
 - (ii) Point P lies inside the shell ($r < R$). [7M]
4. a) Differentiate PIN photodiode from avalanche photodiode. [2M]
- b) Demonstrate the relationship between fill-factor and efficiency of a solar cell through V-I characteristics of a solar cell. [4M]
- b) A solar cell is exposed to light of an input power of 400 mW. The cell area is 2 cm^2 . Calculate its conversion efficiency. Given short-circuit current is 250 mA, fill-factor of a solar cell is 0.4 and open circuit-voltage is 0.55V respectively. [4M]

SECTION - B

Answer all Two questions

2×20M=40M

5. a) Why visible light is not useful for the study of crystal structures? Obtain an expression for Bragg's law of X-ray diffraction.[6M]
- b) The metal niobium (Nb) has a BCC crystal structure. If the angle of diffraction for the (211) set of planes occurs at 75.99° (first-order reflection) when monochromatic x-radiation having a wavelength of 0.1659 nm is used, compute the following: (a) The inter-planar spacing for this set of planes (b) The atomic radius for the Nb atom.[6M]
- c) Solve the Schrodinger's wave equation for a particle confined to an infinite potential box of width L for the energies in various levels and show that the energy is quantized. [8M]
6. a) Outline the expression for magnetic field as per Biot-Savart' law and differentiate it from Ampere's law.[6M]
- b) Calculate the magnetic field using Biot-Savart's law on the axis of a circular coil of radius 0.4m, with 50 turns and carrying a steady current 1.5 Amp at a point P which is at a distance 0.2m from the center of the coil. (Given $\mu_0 = 4\pi \times 10^{-7} \text{ Tm A}^{-1}$)[4M]
- c) Differentiate between LED and diode laser with respect to construction and working. [6M]
- d) Determine the wavelength of light emitted from LED which is made up of semiconductor whose forbidden energy gap is 1.875 eV. Mention the color of the light emitted (Given $h = 6.6 \times 10^{-34} \text{ J-s}$). [4M]